TAHSIN BINTA TOWHID

Computer Science & Engineering Graduate | Software Engineering • Backend • Database • QA

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PROFESSIONAL SUMMARY

Computer Science and Engineering graduate from **BRAC University** with a **3.69/4.00 CGPA** and hands-on experience developing Python-based software, database-driven applications, hardware-integrated systems, and machine-learning pipelines. Strong foundation in **software engineering, backend development, relational database design, API integration, testing, and system analysis**. Experienced in applying Python across software development, computational research, IoT, embedded-system integration, and computer graphics.

TECHNICAL SKILLS

Programming: Python, SQL, C, C/C++, JavaScript, HTML5, CSS3
Python & Data: NumPy, Pandas, Scikit-Learn, Google Colab
Databases: MySQL, Relational Database Design, 3NF Normalization, ACID Transactions, CRUD
Software Engineering: REST APIs, SDLC, SQA, Test Case Design, System & Integration Testing
Tools: Git, GitHub, VS Code, Postman
Hardware & IoT: Arduino, RFID, Sensors, Relays, LCD, Keypad, Serial Communication
Graphics: OpenGL

PROJECTS

Undergraduate Thesis — A Biophysical Fusion Framework for Species-Conditioned Antibacterial MIC Prediction
Python • Scikit-Learn • Computational Biophysics • Feature Engineering • Google Colab
• Developed a multimodal predictive framework combining molecular biophysical descriptors with species-specific genomic embeddings to estimate antibacterial Minimum Inhibitory Concentration (MIC).
• Built feature-extraction and preprocessing pipelines for high-dimensional biological data and multicollinearity reduction.
• Evaluated and benchmarked predictive models using model-error metrics.

Travel Booking System — Software Engineering Project
Python • MySQL • HTML5/CSS3 • REST APIs • Git
• Developed an integrated travel-booking platform for managing destinations, accommodations, and transportation options.
• Implemented booking workflows, dynamic pricing, and mock payment-gateway integration.
• Designed system and integration test cases to validate booking, checkout, and payment workflows.
• Applied software-engineering practices including modular design, testing, and iterative development.

Hotel Management & Reservation System — Database Project
Python • MySQL • Relational Database Design • Git
• Designed and implemented a **3NF relational database** supporting Admin and Customer roles with role-based access control.
• Developed reservation conflict-prevention logic, customer discount functionality, automated billing, and receipt generation.
• Implemented CRUD operations and staff activity logging across core hotel-management workflows.

Smart Library Management & Automation System
Python • Arduino • RFID • LCD • Keypad • Serial Communication
• Developed an automated library-management prototype integrating RFID scanning, keypad input, and LCD-based user interaction.
• Built a Python-based PC application to communicate with Arduino through serial communication for real-time checkout and inventory operations.
• Implemented automated overdue-alert functionality and synchronization between physical hardware interactions and software records.

Automated Planter / IoT Monitoring System
Python • Arduino • Soil Moisture Sensors • Relays • Serial Communication
• Developed an automated plant-care system integrating soil-moisture and environmental sensors with Arduino-controlled irrigation hardware.
• Built a Python-based PC interface to monitor sensor readings and communicate with the Arduino in real time.
• Implemented threshold-based watering logic and documented the system using circuit schematics, hardware state-machine diagrams, and functional specifications.

2D Car Racing Game — Computer Graphics Project
Python • OpenGL • Computer Graphics
• Developed an interactive 2D car racing game using Python and OpenGL with real-time rendering and keyboard-controlled gameplay.
• Implemented vehicle movement, graphical object rendering, collision detection, and game-state logic.
• Applied geometric transformations, animation, and interactive 2D scene-rendering techniques.

EDUCATION

BRAC University, Dhaka, Bangladesh
Bachelor of Science in Computer Science and Engineering | September 2026 | **CGPA: 3.69/4.00**
Relevant Coursework: Database Management Systems, Software Engineering, Microprocessors, Object-Oriented Programming, Data Structures

Milestone College, Uttara, Dhaka
Higher Secondary Certificate (HSC), Science | 2020 | **GPA: 5.00/5.00**

Joydebpur Govt. Girls' High School, Gazipur
Secondary School Certificate (SSC), Science | 2018 | **GPA: 5.00/5.00**

LANGUAGES & COMMUNICATION

Languages: Bangla (Native) • English (Professional Working Proficiency) • Hindi (Conversational) • French (Basic)
Communication: Public Speaking • Debate • Program Anchoring